7 **(a)** A ball of mass 320 g is projected at an angle by a ball launcher. Air resistance is not negligible. The ball is launched from a height of 0.50 m above the level ground.

The variation with time t of the vertical velocity component v_y of the ball is shown in Fig. 7.1.

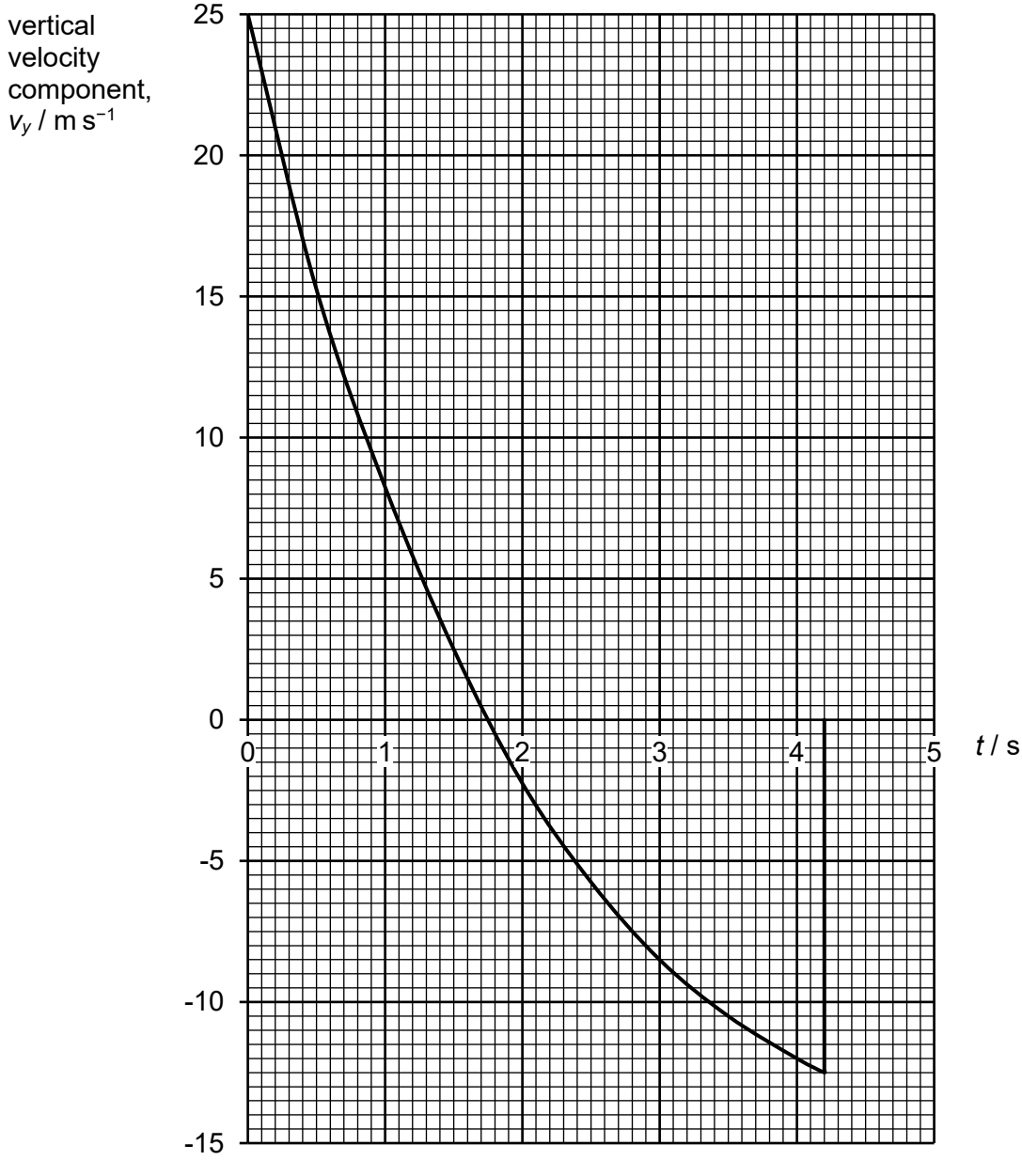

Fig. 7.1

(i) Estimate the vertical height from level ground reached by the ball at t = 1.0 s.

height = …… m [2]

(ii) Estimate the vertical component of the air resistance that is acting on the ball at t = 1.0 s.

air resistance = ……N [3]

(iii) In the case that air resistance is negligible, the ball has a horizontal velocity component of 12.0 m s^{-1}.

1. On Fig. 7.1, sketch a graph to show the variation with time *t* of the velocity vertical component of the ball. Label this line Z. [2]

2. Hence, determine the resultant velocity of the ball at t = 1.0 s.

velocity = …….. m s^{-1} [2]

(b) Another 2 balls, X and Y are travelling towards each other along a horizontal frictionless surface.

(i) Show how the principle of conservation of linear momentum can be derived using Newton's Laws.

……

……

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……

……

…… [3]

(ii) **1.** The two balls collided and rebounded in opposite directions. Table 7.1 shows data for X and Y during the collision.

Table 7.1

ball	mass / g	velocity just before collision / m s^{-1}	velocity just after collision / m s^{-1}
X	50	+4.5	-1.8
Y	M	-2.8	+1.4

The positive direction is horizontal and to the right.

Use your answer in **(b)(i)** to determine the mass of Y.

mass of Y = …… kg [3]

2. State and explain quantitatively whether the collision is elastic.

.....

..... [2]

(iii) The variation of the force that X exerts on Y with time is shown in Fig. 7.2.

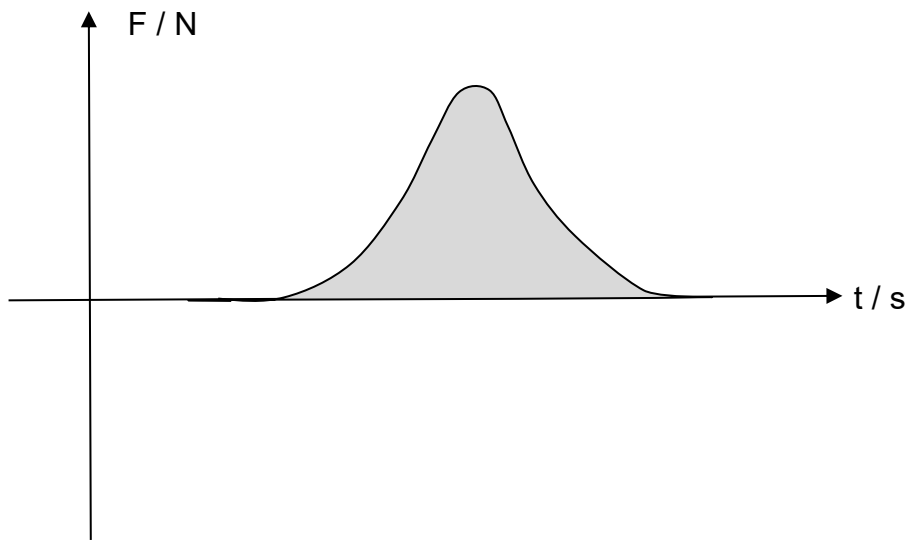

Fig. 7.2

1. Sketch a graph on Fig. 7.2 to show the variation of the force that Y exerts on X with time. Explain your answer.

.....

..... [2]

2. Explain what the shaded area in Fig. 7.2 represents.

.....

..... [1]